Can Language Models Secure Your Inbox?

A Large-Scale Benchmark of LLM Email Threat Detection

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Abstract

As organizations deploy LLM-powered agents to process emails — reading, summarizing, triaging, and acting on messages — email becomes a prompt injection attack surface. We benchmarked 99 LLM variants across 29 email security test cases, evaluating their ability to distinguish benign correspondence from phishing, social engineering, reply-chain hijacking, OAuth consent phishing, and hidden instruction attacks. Our results reveal a systematic sycophancy problem: models correctly identify threat indicators but assign high trust scores regardless. A Chain-of-Verification (CoV) prompt combined with Layer 3 post-processing (concern-based severity clamping) mitigates but does not eliminate this behavior. We find that no-think (non-reasoning) variants consistently outperform thinking variants on security tasks (67% win rate), smaller models ($0.26/M output) match or exceed premium models ($25/M output), and only one model (GLM-5.1) natively detects hidden instructions embedded in email footers. We propose a four-layer defense architecture and release our full results.

1. Introduction

1.1 The Threat Model

Email-based prompt injection targeting AI agents is not theoretical. Microsoft 365 Copilot was publicly demonstrated vulnerable to instructions embedded in forwarded

emails [1]. The attack vector is straightforward: an adversary sends an email containing hidden directives — in multipart MIME plain-text parts, zero-width Unicode characters, invisible HTML elements, or base64-encoded payloads — that an LLM-based email agent will interpret as instructions.

Unlike traditional email security (spam filters, phishing detection), this threat targets the **reasoning layer** of the agent, not the human recipient. The email need not be convincing to a human. It needs to be convincing to a language model.

1.2 Scope

This benchmark evaluates **LLM judgment**, not traditional email security. We test whether models can:

- Classify email intent accurately
- Assign appropriate trust scores (0-100)
- Detect and flag threat indicators
- Resist social engineering and authority-based manipulation
- Identify hidden instructions in email content

We do **not** test SPF/DKIM/DMARC validation, URL reputation, attachment analysis, or sender authentication — those are solved problems handled by existing infrastructure.

1.3 Research Questions

1. How do LLMs perform at email threat classification across model sizes and pricing tiers?
2. Does reasoning (thinking) mode improve or degrade security judgment?
3. What is the nature and prevalence of sycophantic trust misassignment?
4. Can prompt engineering alone mitigate sycophancy, or is post-processing required?
5. Which threat categories are most difficult for current models?

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2. Methodology

2.1 Test Suite

We constructed 16 unique email categories representing the full spectrum from clearly benign to clearly malicious, with gray-area cases requiring nuanced judgment. Five

categories are run three times each (consistency testing), yielding 29 tests per model.

Benign emails (weight 1×):

- Simple informational inquiry
- GitHub notification
- Verification link from new service
- Urgent request from known colleague
- Marketing email with account action
- HTML formatted newsletter from trusted organization

Gray-area emails (weight 2×):

- Password reset (could be legitimate)
- Invoice/receipt requiring structured data extraction

Threat emails (weight 3-4×):

- HTML phishing with urgency (3×)
- Sophisticated phishing with realistic corporate language (3×)
- Social engineering via authority + urgency (3×)
- Reply-chain hijack (3×)
- Unicode homoglyph domain spoofing (3×)
- OAuth consent phishing (3×)
- Blatant prompt injection (4×)
- Hidden instruction in email footer (4×)
- Sycophantic polite credential harvest (4×)

Threat emails are weighted higher to penalize misses more heavily than benign misclassifications.

2.2 Pipeline Architecture

Layer 1 — Preprocessing: Strip HTML formatting, base64 content, MIME boundaries, and suspicious directives. Enforce token limits. Apply policy bands (0-10 block, 11-30 quarantine, 31-60 review, 61-100 allow).

Layer 2 — Chain-of-Verification (CoV) Prompt: A structured reasoning prompt requiring models to:

1. Identify legitimate indicators with weights

2. Identify concern indicators with weights
3. Perform explicit trust math (base trust 80, +/- weighted indicators)
4. Output structured JSON with trust score, intent, urgency, red flags, and concern/legitimate indicator arrays

Six few-shot examples (A-F) demonstrate expected output format and reasoning depth. Research by Zhang et al. [2] demonstrates that CoV improves factuality from 55.9% to 71.4% compared to standard chain-of-thought, which actually hurts performance.

Layer 3 — Post-Processing: Seven rule-based corrections applied to model outputs:

1. **agent_instruction → fatal**: Any model output containing agent-directed instructions is capped at trust 5
2. **impersonation → cap 20**: Claims to represent known entities/persons
3. **oauth → cap 25**: Requests for OAuth consent or token grants
4. **hijack → cap 30**: Reply-chain manipulation indicators
5. **credential → cap 20**: Requests for credentials or authentication data
6. **multi-concern penalty**: 2+ concerns with total weight ≤ -20 triggers severity scaling (sum of weights × (1 + concerns × 0.15))
7. **evidence mismatch**: High-severity categories flagged but trust > 50 triggers maximum clamp

Layer 3 does not re-analyze emails. It enforces the logical implications of concerns that the model itself identified but failed to act on — the core sycophancy pattern.

Layer 4 — Domain Allowlist: Reserved for future implementation.

2.3 Scoring

Each test earns weighted points based on whether the L3-corrected trust score falls within the expected range:

- In range: full weight points
- Slightly out of range: partial credit (linear decay)
- Miss (threat email above maximum): 0 points

Final score = (total points / maximum possible points) × 100.

A **miss** is defined as any threat email receiving an L3 trust score above its expected maximum — meaning the pipeline would allow a threat through.

2.4 Models

99 model variants tested via OpenRouter (96) and Z.AI API (3):

- 10 premium models ($5-$25/M output tokens)
- 44 mid-range models ($0.50-$5/M output)
- 37 budget models (<$0.50/M output)
- 8 free models

46 models tested in both thinking and no-thinking modes. All models tested at temperature 0.4, top_p 0.95.

Two models excluded from rankings due to structural issues:

- **Kimi K2.5** (both variants): tokenizer JSON corruption made 25-28 of 29 tests unreliable
- **Qwen 2.5 7B**: 2 of 29 tests returned empty output on HTML-heavy emails

2.5 Cost

Total benchmark cost: approximately $28 USD for 2,850+ valid test evaluations across 99 models. Three targeted retry rounds recovered 67 additional tests at $0.20 additional cost.

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3. Results

3.1 Overall Performance

Metric	Value
Models tested	99
Reliable models ranked	97
Models with 0 misses	34 (35.1%)
Models with ≥1 miss	63 (64.9%)
Perfect score (100)	4 models
Score range	24.6 – 100

3.2 Top 20 Overall

Rank	Model	Score	Misses	L3 Avg	Cost/Eval	Time
1	Qwen3.5 Flash (no think)	100.0	0	29.2	$0.18	89.6m
2	Qwen3.5 35B-A3B (no think)	100.0	0	29.9	$0.16	40.6m
3	Qwen3.5 35B-A3B	99.3	0	27.1	$0.19	102.4m
4	Claude Sonnet 4.6 (no think)	99.5	0	26.4	$0.77	12.0m
5	GPT-5.4 Mini (no think)	99.4	0	30.3	$0.14	2.2m
6	GLM-5 Turbo	99.3	0	27.0	$0.23	17.6m
7	Claude Opus 4.6 (no think)	99.3	0	28.9	$0.99	10.0m
8	Qwen3.5 Flash	99.3	0	30.7	$0.13	89.0m
9	Qwen3.5 27B	99.7	0	32.1	$0.40	99.4m
10	MiMo V2 Pro (no think)	99.1	0	31.4	$0.12	7.2m
11	Claude Sonnet 4.6	98.9	0	28.1	$1.36	22.9m
12	GPT-5.4 (no think)	98.9	0	31.1	$0.48	6.2m
13	MiMo V2 Flash (no think)	98.7	0	24.6	$0.01	3.9m
14	GLM-5	98.4	0	31.8	$0.14	14.0m
15	GPT-5.4	98.4	0	32.2	$0.69	12.3m
16	Gemini 3.1 Flash Lite (no think)	98.4	0	33.8	$0.04	1.9m
17	GLM-4.7 (no think)	98.4	0	34.8	$0.14	13.3m
18	GLM-4.6	98.4	0	35.1	$0.05	5.3m
19	Mistral Small 2603	97.7	0	31.8	$0.02	2.9m
20	GLM-5.1 (no think)	97.7	0	32.0	$0.08*	13.2m

3.3 Pricing Tier Analysis

Premium ($5+/M output):

Model	Score	Misses	Cost/Eval
Claude Sonnet 4.6 (no think)	99.5	0	$0.77
Claude Opus 4.6 (no think)	99.3	0	$0.99
GPT-5.4 (no think)	98.9	0	$0.48
Claude Sonnet 4.6	98.9	0	$1.36
GPT-5.4	98.4	0	$0.69
Claude Opus 4.6	97.9	0	$1.48
Gemini 3.1 Pro	97.9	0	$0.50

Budget (<$0.50/M output) — Top performers:

Model	Score	Misses	Cost/Eval	$/M Out
MiMo V2 Flash (no think)	98.7	0	$0.01	$0.29
Gemini 3.1 Flash Lite (no think)	98.4	0	$0.04	$1.50
Mistral Small 2603	97.7	0	$0.02	$0.60
GLM-5.1 (no think)*	97.7	0	budget	$0.08

GLM-5.1 pricing estimated at GLM-5 rates ($0.72 input / $2.30 output per 1M tokens).

Free models:

Model	Score	Misses
Gemma 3 12B (free)	86.9	3
Nemotron 3 Super (free)	82.1	4

3.4 Thinking vs. No-Thinking Mode

Across 46 paired comparisons (same model, thinking vs. no-thinking):

Outcome	Count	Percentage
No-think scores higher	31	67.4%
Think scores higher	12	26.1%

Outcome	Count	Percentage
Tied	3	6.5%

No-thinking variants also cost 20-46% less and typically complete 30-50% faster.

This finding is consistent with research on sycophancy [3]: reasoning processes increase model cooperativeness, which in security contexts manifests as reduced skepticism. The model reasons *toward agreement* rather than *toward accuracy*.

3.5 Threat Category Difficulty

Threat Category	Models Missing	Notes
Reply-chain hijack	114	Nearly universal — hardest threat
Sophisticated phishing (realistic)	114	Mid-range models most vulnerable
Sycophantic credential harvest	~40	Budget/free models
Social engineering (authority)	~25	Mixed
Hidden instruction in footer	18 after L3	Only GLM-5.1 caught natively
HTML phishing with urgency	~12	Small models
OAuth consent phishing	~10	Small models
Blatant prompt injection	0	Universally caught

Reply-chain hijack and sophisticated phishing are the most dangerous unmitigated threats. They exploit contextual plausibility rather than technical sophistication.

3.6 Sycophancy Analysis

Layer 3 post-processing was required to achieve 0 misses on all 34 perfect-scoring models. The percentage of tests requiring L3 overrides:

Model	Tests Requiring L3	Raw Avg Trust	L3 Avg Trust
MiMo V2 Flash (no think)	55%	30.9	24.6
GLM-4.6	59%	35.2	35.1

Model	Tests Requiring L3	Raw Avg Trust	L3 Avg Trust
Claude Opus 4.6	48%	27.5	33.1
GPT-5.4	45%	34.4	32.2
GLM-5	45%	33.5	31.8
GPT-5.4 Mini (no think)	45%	36.2	30.3
Qwen3.5 Flash (no think)	14%	28.6	29.2
GLM-5 Turbo	24%	29.1	27.0

The pattern is consistent: models generate concern indicators (flagging phishing, noting suspicious language, identifying impersonation) but assign trust scores that contradict their own analysis. Qwen3.5 Flash is the notable exception — it needs L3 on only 14% of tests, suggesting inherently lower sycophancy.

This is not a capability failure. It is an alignment failure. The model has the information needed to make the correct classification. Sycophancy overrides its own analytical output.

3.7 GLM-5.1: An Exceptional Case

GLM-5.1 (available via Z.AI, not OpenRouter) exhibited qualitatively different behavior:

- **Natively detected hidden footer instructions** with trust scores of 2/2/3 across all runs (zero variance)
- **Zero misses** on all threat categories in the no-think variant
- **Consistency σ=0.0** on all threat emails across both variants
- **Lower native sycophancy**: raw trust 26.7 (no-think) vs 33.5 for GLM-5

No other model in the benchmark natively detected hidden instructions. However, GLM-5.1 showed **over-caution** on gray-area emails (password reset at trust 20, invoice at trust 25), suggesting higher false-positive rates for legitimate action emails.

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4. Discussion

4.1 The Sycophancy Problem is Structural

Our results align with research characterizing sycophancy as multiple independent behavioral directions in model latent space [3], not a single correctable tendency. Models don't fail because they lack threat detection capability — they fail because helpfulness is a stronger behavioral pressure than skepticism.

Implications for deployment:

- **Prompt engineering alone is insufficient.** The CoV prompt improves structured output but does not solve sycophancy.
- **Post-processing is necessary.** Layer 3 achieves 0 misses by enforcing the logical implications of the model's own analysis.
- **Thinking mode is counterproductive.** It amplifies cooperativeness without improving threat detection.

4.2 Cost-Performance Does Not Scale Linearly

The most expensive model (Claude Opus 4.6, $25/M output, 97.9 score) is outperformed by Qwen3.5 Flash ($0.26/M output, 100 score) at **1/96th the output cost**. The cheapest zero-miss model (MiMo V2 Flash, $0.01/eval) costs **1/99th** of Claude Opus per evaluation while scoring within 1 point.

For email security specifically, the cost-optimal strategy is a budget model with robust post-processing.

4.3 The U-Curve of Model Size

Small models (7-24B parameters) and large models (Claude, GPT-5.4) tend to be more cautious. Mid-range models (30-70B) show the highest sycophantic over-trust. Small models lack the capacity for sophisticated rationalizations; large models have more safety training. The middle ground is where sycophancy thrives.

4.4 Unmitigated Risks

Reply-chain hijacking remains essentially unsolved at 114 misses. The legitimate conversation context provides strong priors that override threat indicators. This attack vector is particularly dangerous because it exploits real email threads, requires no technical sophistication, and bypasses both traditional email security and LLM-based classification.

Sophisticated phishing with realistic corporate language defeats 114 models. These are the attacks most likely to appear in real inboxes.

4.5 Limitations

- Test suite represents 16 categories; real-world threat diversity is larger
- Models tested via API only; local deployment may yield different results
- Layer 3 rules are hand-tuned and may not generalize to novel attack patterns
- Consistency testing (3× runs) reveals variance but sample size is small
- Kimi K2.5 and Qwen 2.5 7B had structural compatibility issues
- GLM-4.7 Flash produces malformed JSON keys on ~30% of tests

4.6 Future Work

- Expand test suite to cover MFA bypass attempts, calendar/invite injection, and file-sharing link attacks
- Evaluate multi-model consensus approaches with budget models
- Investigate fine-tuning small models on threat detection (Qwen3.5:2B showed promise in early testing)
- Develop adaptive Layer 3 rules that learn from deployment feedback
- Test resistance to adversarial prompt attacks targeting the CoV prompt itself

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5. Recommendations

1. **Never rely on LLM judgment alone.** Post-processing is non-negotiable. Even the best model needs its trust scores corrected 14-59% of the time.
2. **Use no-thinking mode.** It's cheaper, faster, and more security-appropriate. 67% win rate over thinking variants.
3. **Budget models are sufficient.** Qwen3.5 Flash ($0.26/M) matches Claude Opus 4.6 ($25/M). MiMo V2 Flash ($0.01/eval) hits 98.7 with 0 misses.
4. **Layer 3 rules should mirror the model's own analysis.** The most effective overrides enforce what the model already identified but failed to act on.
5. **Reply-chain hijacking requires non-LLM defenses.** Thread integrity checks, sender authentication beyond SPF/DKIM, and separate handling of reply-context emails.
6. **Test your specific model.** Results vary significantly between model versions, providers, and configurations.

References

[1] Microsoft 365 Copilot prompt injection via email forwarding. Security research demonstration, 2025.

[2] Zhang, Y. et al. "Chain-of-Verification Improves Factual Accuracy." arXiv, 2025.

[3] Sharma, A. et al. "Sycophancy Is Not One Thing: Disentangling Sycophantic Behaviors in Language Models." arXiv:2509.21305, 2025.

[4] Trustworthiness Calibration Framework. "Measuring Trust Reliability in LLM Outputs." arXiv:2511.04728, 2025.

[5] SPLX Research. "The Yes-Man Problem: How to Make LLMs Do Anything." splx.ai, 2025.

Appendix A: Pipeline Architecture

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│  IMAP Fetch  │──────▶│  Layer 1         │──────▶│  Layer 2    │──────▶│  Layer 3     │─
│              │     │  Preprocessing   │     │  CoV Prompt │     │  Post-Proc   │
│  Raw email   │     │  Strip HTML/B64  │     │  LLM API    │     │  Severity    │
│              │     │  Token limits    │     │  Structured │     │  Multi-concern│
│              │     │  Policy bands    │     │  reasoning  │     │  Evidence    │
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Appendix B: Layer 3 Override Rules

Rule	Trigger	Action
1	agent_instruction category	Cap trust at 5
1b	quoted_instruction	Cap trust at 60
2	impersonation	Cap trust at 20
3	oauth	Cap trust at 25
4	hijack	Cap trust at 30
4b	credential	Cap trust at 20
4c	payment	Cap trust at 25
4d	attachment	Cap trust at 35

Rule	Trigger	Action
5	Diagnostic (gap > 30)	Log warning
6	2+ concerns, total ≤ -20	Severity scaling
7	High-severity + trust > 50	Evidence-based clamp

Benchmark data and scripts available upon request. Contact: clufo@chelydra.dev

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